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Comparing users' performance and game experience between a competitive and collaborative brain-computer interface

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ABSTRACT

The recent advancements in Brain-Computer Interfaces (BCIs) technology made possible for its employment in entertainment applications, such as brain-controlled games. However, there are still many challenges to overcome in the design and development of such applications, especially when considering the interaction between multiple users. This paper presents the results of a controlled experiment aimed at comparing the performance and game experience of both competitive and collaborative matches in a multiplayer BCI game. In the experiment, sixteen subjects were divided in pairs into two groups, and each pair played a total of five competitive or collaborative matches of the attention-based game. Quantitative and qualitative data were collected to investigate key features of the subjects' interaction with both the BCI platform and the other players. It was found that there is no statistically significant difference in performance between the subjects of both groups, and between multiplayer and single player modes. Although the group playing competitively reported overall better scores for game experience, there was no statistically significant difference between groups. There were significant correlations between performance and positive aspects of game experience for both modes, and multiplayer matches were significantly shorter than single player ones, although this effect could not be attributed to the mode of interaction alone.

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BCI; EEG; HCI; game; neurofeedback

1. Introduction

Brain-Computer Interface (BCI) systems are a novel kind of human-computer interface that enables the machine to read, interpret and translate the user's brain signals and intentions as commands for specialised applications (Graimann, Allison, and Pfurtscheller 2009). The great advantage of BCI systems is that they allow any person to use it, as only cerebral activity is required to control the application (Sedeeq and Romano 2014). This makes BCI a powerful alternative for patients suffering from any physical impairment that would hinder the interaction between conventional human-computer interfaces (Ferreira et al. 2013). With the increasing advance of BCI technology, especially those based on electroencephalography (EEG) – a non-invasive method popular for its convenience and cost (Marshall et al. 2013) –, BCI systems are starting to become accessible for consumer-grade applications (Gürkök and Nijholt 2012; Nijholt 2012; Vasiljevic and de Miranda 2020b).

While traditional BCI research has been focussed on analysing and improving the overall performance of the system itself, recent works are starting to investigate the

applications of BCI for entertainment purposes – not only for unhealthy subjects, but for healthy users as well (Obbink et al. 2012; Wong et al. 2015; Vasiljevic and de Miranda 2020a; Nijholt 2015b). The emergence of low-cost, consumer-grade devices has allowed for even more research opportunities in the field of Human-Computer Interaction (HCI), as these entertainment applications, such as games, can now be developed targeting the general population, instead of exclusively for laboratory research and medical purposes (Bos et al. 2010).

However, given that there are a number of challenges in the design and development of BCI-based games (Ferreira et al. 2014; Vasiljevic, de Miranda, and de Menezes 2018b), these challenges are yet augmented in the process of developing a game that also considers multiplayer interaction. Not only must the developer consider the typical implementation hinders of a multiplayer game, but must also take into consideration the limitations of the BCI platform itself, such as the employed equipment, the brain waves processing algorithm, the natural and biological oscillation of the captured physiological signal, and how this signal may be

influenced by external stimuli (Vasiljevic and de Miranda 2019a, 2019b).

In addition, given the social aspect added by the multiplayer interaction, the way that the players interact may greatly influence the game experience and the players' performance, as the social interaction itself may possibly change the captured control signal (Obbink et al. 2012). The investigation of these social factors that may interfere with the player's game experience could help designing future multiplayer neurofeedback-based games focussed on either of both modes of social interaction, and may help to comprehend the influence of actively interacting with other people in the subject's level of attention, which is particularly important for games based on neurofeedback.

Past studies have analysed a number of factors regarding multiplayer interaction in BCI and non-BCI games, including possible effects of game mode (competitive or collaborative) and number of players (single-user condition versus multi-user condition) on both game performance and game experience. Although there is evidence that suggest that game mode has little effect on player performance, there is usually a significant difference in both experience and performance when comparing single and multiplayer conditions, with subjects preferring and performing better on multiplayer conditions. However, considering active BCI games (i.e. those in which the players actively modulates their brain waves to generate a command), both the performance and game experience may also be affected by the employed EEG control signal, especially those based on neurofeedback, in which the player must focus on concentrating or relaxing in order to modulate their brain waves. There is currently a small body of studies that investigate these factors in multiplayer BCI games, employing different types of games, control signals, performance metrics and qualitative instruments, making it difficult to objectively compare their results.

In this context, this paper presents and describe a controlled experiment using a BCI-based multiplayer game. This experiment allowed analysing both qualitative and quantitative aspects of the subjects' interaction with the BCI platform, with the game itself, and with the other player, focussing on the influence of the social interaction and the multiplayer game mode (i.e. competitive or collaborative) on the player's performance and game experience. Based on these objectives, the experiment was guided by the following Research Questions (RQs):

- **RQ1:** Is there a difference in performance between subjects playing collaboratively and competitively?

- **RQ2:** Is there a difference in game experience between subjects playing collaboratively and competitively?
- **RQ3:** Is there a relation between performance and game experience in both groups?
- **RQ4:** What are the subjective or qualitative differences in game experience between groups?

These research questions aim at investigating how the relation between those factors could impact in the players' experience while playing the game and, consequently, in the design and development of multiplayer BCI games. Although there has been efforts to understand how these social aspects may affect various factors regarding the performance of the players and their interaction with the game, the relations between social interaction mode, attention and game experience for the design of such games, especially in the context of entertainment, are still largely unknown.

For answering the RQs, the experiment was performed with 16 subjects, which were randomly divided into two groups of equal size, with each of these groups playing collaboratively or competitively. Each pair of subjects played five game matches of the same mode: competitively, against each other, or collaboratively, against the computer. Quantitative and qualitative data were employed to compare the performance and game experience between the two groups. Based on the results of the experiment, we were able to identify the relations between changes in attention and social interaction, which could be employed to aid in the design and development of future BCI games based on multiplayer interaction. In addition, we also analyse the differences in match duration between the two games modes, and between single and multiplayer modes based on previous experiments.

Thus, the main contributions of this study are two-fold. First, it builds upon the current investigation of the differences, both in performance and game experience, between playing neurofeedback-based BCI games with different modes of social interaction, while also considering the influence of time and number of matches in the subjects' performance. Second, it investigates how these factors may influence on the interaction design of such games, which is particularly important for the development of future systems that take these factors into consideration. We compare the design of the game employed in the current study to those used in related works, especially considering the differences in EEG-based control signals and how the required mental strategy (or strategies induced by the design or respective game mechanics) may influence on the subjects' performance and experience.

This paper is organised as follows: Section 2 shows the related work in the area; Section 3 describes the research method employed in this study; in Section 4 the quantitative and qualitative results of the experiment are described; in Section 5 the results are discussed; and Section 6 concludes the paper.

2. Related work

Past studies have investigated the possible differences of competitive or cooperative/collaborative play in the context of a BCI-based game in different conditions. Salminen et al. (2009), for example, analysed the frontal EEG asymmetry and the game experience (measured using the GEQ – Game Experience Questionnaire) of 24 pairs of players during both modes of social interaction, but found no significant differences between the playing modes. Bonnet, Lotte, and Lécuyer (2013) evaluated the performance and subjective experience of both single and multiuser conditions in a BCI game. Two experiments were performed: one consisted of 10 pairs of subjects in a collaborative task, while the other consisted of four pairs of subjects in both competitive and collaborative tasks. The experiments revealed that, although the multiuser condition was preferred over the single user condition, there was no significant preference over competitive or collaborative tasks, nor significant difference in the subjects' performance between the two tasks, which is in accordance with the results obtained from (Salminen et al. 2009).

In a similar manner, Sekhavat (2020) presents a multiplayer EEG-based game, which uses the attention value from a NeuroSky Mindwave EEG device to adjust various parameters in the game. The game can be played either competitively or collaboratively, and was employed to verify differences in performance and game experience in both modes and in comparison to standard positive/negative reinforcement conditions. The study showed that there was a significant difference in performance and game experience (measured using the GEQ) between playing competitive/collaborative in comparison to the standard positive/negative reinforcement conditions. However, there was no significant difference between the competitive and collaborative conditions.

Obbink et al. (2012) compared the changes in the social interaction between using a traditional interface (mouse) and a BCI in a cooperative game. The researchers found that the 10 pairs of participants cooperated better while using the traditional interface, indicating that the BCI control may have hindered the cooperative interaction between the players. In addition, Spapé et al. (2013) found that there is a higher concordance between the EEG signals (i.e. similar patterns of brain activity) of

the participants while playing a game competitively, in contrast to playing cooperatively. However, Valeriani, Poli, and Cinel (2016) demonstrated that collaborative BCIs can be used to significantly improve the system's accuracy and enhance group perception.

In the work of Shenjie et al. (2014), an attention-based two-player game is proposed and evaluated over three days with four subjects. The evaluation showed that the subjects not only were able to sustain their attention longer over the experiment sessions, but also increase their performance with each consecutive day. Cruz et al. (2018), on the other hand, presented a cooperative SSVEP¹-based two-player game focussed on the user experience and game design. The authors demonstrated that, even with low BCI performance, the twelve subjects that participated in the study were able to enjoy the game and its social aspect, showing the importance of good game design in BCI-based games.

The focus of our work, in this context, is the difference, both quantitative and qualitative, between competitive and collaborative BCI gameplay and its consequences on the interaction of multiplayer EEG-controlled games, especially those focussed on entertainment purposes. We aim to investigate how the performance and game experience differs from these two perspectives using a BCI-based competitive/collaborative game controlled by the subjects' attention. As stated by Nijholt and Poel (2016), there is still a need for further research in the attempt to model various forms of multi-brain interactions, as to allow a more systematic approach to designing future multi-brain BCI applications. We seek to investigate not only the objective influence of the game modes on the quantitative variables, but also qualitative aspects of the interaction and the subjective view of the participants on the game and how it is played, as explained next.

3. Method

A controlled experiment methodology (Lazar et al. 2017; Robson 2002; Ross and Morrison 2008; Wohlin et al. 2012) was employed in this study given the deductive nature of the research questions, which aims at trying to find a cause-effect relationship between the performance/game experience and the competitive/collaborative gameplay. Mostly, quantitative methods were employed to analyse the results, although qualitative tools were also employed to complement, explain and corroborate the quantitative methods. This empirical-based methodology is widely employed in the field of HCI, especially in BCI studies (Zhou et al. 2014; Antle et al. 2015; Liu et al. 2015; Jeunet et al. 2018), for investigating causal effects in controlled environments.

The experiment in this study was designed to evaluate the social aspects of a multiplayer neurofeedback game, especially regarding the interaction between players and how this interaction could have affected the performance and game experience of the subjects. This section describes and details the experiment's methodology.

3.1. Research questions

The first research question, RQ1 (defined in Section 1), involves the investigation of a possible difference in performance due to the social interaction during the BCI game, i.e. whether playing competitively or collaboratively significantly impacts the average attention of the players in a neurofeedback-based game. The main hypothesis is that the social presence of other players and the pressure to keep a better performance in comparison to them may disrupt or increase the focussed attention of the subject. Thus, the null hypothesis is that there is no difference in performance between the competitive and the collaborative groups.

Past studies suggest that there might not be a significant difference between the two modes, although there is not sufficient evidence to show that this holds true in the majority of cases. Nevertheless, directly comparing players' performance in both modes provide an objective parameter to serve as base for other quantitative and qualitative analysis. Even when no difference in performance is found between groups, the subjective quality of the user experience while playing may be analysed in relation to how the users perceived their performance (measured in terms of attention, given the game mechanics), and if this experience in terms of performance is different depending on the game mode.

The second research question, RQ2, is aimed at investigating whether there is a difference in the overall game experience while comparing both modes of social interaction. In this case, the main hypothesis is that the player may perceive a better or worse game experience depending on the mode they are playing, especially in relation to the social presence of the other player and how they are interacting, i.e. competitively or collaboratively. The null hypothesis, in this case, is that there is no difference in the game experience between the two groups.

The third research question, RQ3, regards the relationship between the factors investigated in RQ1 and RQ2, i.e. how does the game performance, measured as the subjects' attention, is related to their game experience. In this case, we aim to investigate if a better or worse performance is related to a better or worse game experience, and if this relation holds

when considering the collaborative or competitive game modes (e.g. if a worse performance could still yield a good game experience if playing collaboratively or competitively).

This RQ focus on the hypothesis that the game experience may not necessarily be affected by the user's performance, e.g. even with a bad performance, players may consider the game experience to be positive, and vice-versa. As stated in Section 2, related studies report that it is possible to achieve a positive game experience even with low performance, although different scenarios were employed, i.e. a collaborative-only game using a different control signal (SSVEP). This could give insights regarding which aspects of the interaction with the game and with other players may have greater impact on the experience other than difficulty alone. The investigation of this RQ also allow to verify which aspects of the game experience (i.e. which dimensions of the GEQ instrument) are directly or indirectly related with performance, including social aspects of user interaction.

Finally, RQ4 is an exploratory question aimed at investigating the qualitative aspects of the subjects' interaction with the game and with the other subjects. We seek to analyse the subjective views of the participants in relation to this interaction, including their opinion on the mode of interaction (competitive or collaborative), the game, on using their brain waves to play, difficulties, and which mental strategies (i.e. to concentrate) they used to play the game. This analysis is related with the goals of RQ3, i.e. understanding which aspects of the interaction may affect game experience in addition to performance, but from a qualitative perspective.

3.2. Participants

The experiment was conducted with sixteen volunteer subjects (S1–S16), being twelve male and four female. Participants ranged from 20 to 32 years ($M = 24.8$, $SD = 2.66$). No participant was clinically diagnosed with attention-deficit disorders nor had physical impairments.

The only selection criteria adopted in this experiment was that the participant could not have any metal implants in the forehead region, since the device reads electrical signals from the brain and such implants could interfere with the captured signals, affecting the accuracy of the device.

3.3. Ambient and materials

The same two computers were used by all subjects in the sessions of the experiment. The experiment was

conducted in a closed and climatized room. The computers were equipped with an Intel Core 2 Duo processor, Windows 7 64-bits operational system, 4 GB RAM, and a 15" monitor, adjusted to the subjects' height. All sessions were recorded with the participants' consent, using four Microsoft LifeCam Cinema cameras connected to two separate computers, avoiding any possible loss of performance in the game. The IBM SPSS Statistics software was employed to perform all statistical tests and data analysis.

The BCI devices employed to acquire the subjects' brain signals were two NeuroSky MindWave, both model MW001. The MindWave is a consumer-grade, low-cost headset device that reads electrical signals from the brain through the scalp (the area bordered by the neck and the face). It uses a single electrode positioned at the Fp1 position in the International 10-20 System (Niedermeyer and Silva 2004), which is located in the forehead, above the left eyebrow. The headset is able to acquire the level of attention of a subject using a set of algorithms called eSense, which filters specific features of the brain signals and translates them into a number that measures how the user is concentrated. This number, received every second from the device, ranges from zero to 100, being directly proportional to how much the user is concentrated (i.e. the higher the subject's attention, the higher the eSense value).

The reliability of the NeuroSky devices as a research tool has been accessed to validate its employment in scientific experiments (Ekandem et al. 2012; Rogers et al. 2016), and the quality of the acquired signals were compared to medical-grade devices (Johnstone, Blackman, and Bruggemann 2012; Ratti et al. 2017). These devices have also been used in a variety of BCI studies, e.g. for educational purposes (Xu and Zhong 2018), and in the context of BCI-based games or gamified applications, such as the works of Ko et al. (2009), Joselli et al. (2014), Antle et al. (2015, 2018, 2019), Li et al. (2021), Park et al. (2018) and Weinell et al. (2015), in which the devices were employed for both evaluation and control. The attention levels provided by the NeuroSky devices have also been evaluated for their reliability and comparison with self-assessed attention values (Rebolledo-Mendez et al. 2009; Chen and Huang 2014; Chen and Wu 2015).

The Game Experience Questionnaire (GEQ) (Poels et al. 2007; Norman 2013) was employed in this experiment for help answering the research questions and to gather more information about the subjects' interaction with the game. The GEQ is divided into three modules, each assessing different components of game experience: the core module, the social presence module,

and the post-game module. The core module assesses game experience based on immersion, flow, competence, positive affect, negative affect, tension and challenge. The social presence module assesses psychological and behavioural involvement of the player with other entities or players. The post-game module assesses how players felt after they had stopped playing. Each module is composed of several items (67 in total), which should be answered using a 5-point unipolar intensity-based scale, with points ranging from 'not at all' (0) to 'extremely' (4). The intensity of the component is calculated as the average of the value of its items. Examples of items are 'I was deeply concentrated in the game', from the core module, and 'My actions depended on the others' actions', from the social presence module. The questionnaire was employed to measure the game experience in a number of biofeedback-based studies, e.g. Salminen et al. (2009), Mühl et al. (2010), Hilborn et al. (2013), Spapé et al. (2013), Wollmann et al. (2016) and Sekhavat (2020).

3.4. Multiplayer attention-based BCI game

Mental War is a BCI-based tug-of-war game, in which the player uses his/her level of attention to pull the rope (Vasiljevic, de Miranda, and de Menezes 2018a). The player that manages to pull the other to the centre of the arena wins the match. The game was developed to be played both individually against the computer (single player, Figure 1(a)), in three different difficulty modes (easy, medium and hard), and in teams (multiplayer). In the multiplayer version, the player can choose to play competitively or collaboratively, together or against other players and/or the computer. Competitively, the player uses his/her attention to compete against other players or against the computer (Figure 1(b)). Collaboratively, the player can join efforts with other players against the opposite team (Figure 1(c)).

The game match is composed of two teams, each on one side of the arena. Each team can be composed of one to three members, and each member can be a human or a machine player (i.e. a real person or the computer – which can be set at any of the three available difficulties of the single player mode). Additionally, it is not necessary that the teams have the same number of players, as the average attention of each member is averaged with the ones from the other members.

The averaged values of each team allows for a more collaborative gameplay, as both members should maintain their average attention high enough so the mean attention is higher than the other team's mean attention. Also, it allows for one single player to be able to compete with two or three other players without penalties, and to

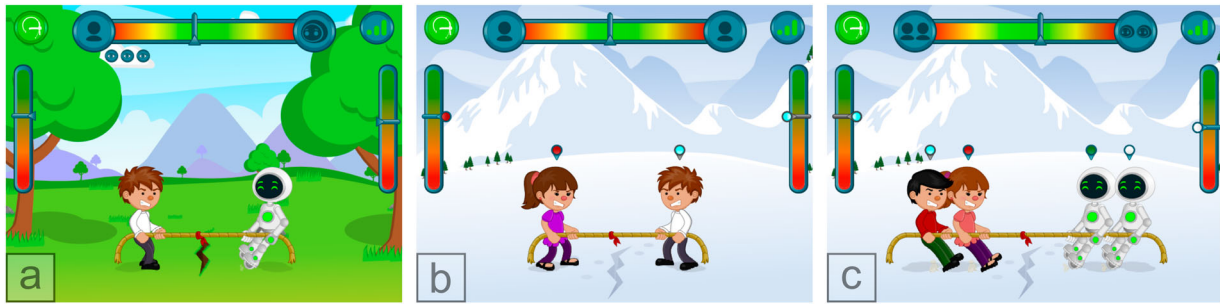


Figure 1. Examples of a (a) single player match, a (b) competitive match and a (c) collaborative match.

create ‘custom’ difficulties by combining two or more machine players of different difficulty levels. In this experiment, however, only the “Easy” difficulty of the machine player was used as default, in which the computer maintains an average attention value between 55 and 65.

The original version of the game uses the average of the player’s attention throughout the entire match as the force value. In order to avoid long match duration due to stagnation of the averaged value (since an attention value is read every second), only the last 20 attention values are considered in the average calculation, i.e. a 20 s rolling average. This creates more variable force values during the match, allowing for the player to recover from a period of low attention, and forcing him/her to maintain the attention high during most of the duration of the match in order to win.

3.5. Procedure

Each subject participated in one session of the experiment, which has one pair of subjects and was recorded with the four digital cameras. In the beginning of the session, the two subjects filled out a demographic form and the experiment was explained by one researcher. Both subjects watched a video that shows the game and how it works with the BCI device, and were free to ask questions at any time. The purpose of the tutorial video was to have the same demonstration of the game for all subjects. The eight pairs of subjects were divided into two groups of equal size: the competitive group, in which each pair of subjects played against each other (Figure 2(b)); and the collaborative group, in which each pair played the game collaboratively to try to defeat the pair of computer players, set in the easy difficulty mode (Figure 2(c)). To assure internal validity, the pairs of subjects were randomly assigned to each group.

Before playing in groups, each subject played an individual, single player match for 30 s in the default easy difficulty (Figure 2(a)). This calibration match

allowed for the subject to familiarise with the game and its controls, and to assure that there was a stable connection with the EEG device. Then, five multiplayer matches were played in the experimental session. The multiplayer matches were not timed, i.e. they only ended when one of the teams was defeated. The number of matches was defined empirically while taking into consideration that each match had a relatively short duration during the initial test phases (below two minutes).

At the end of the session, each player was asked to answer a GEQ form in order to gather more information on the subjects’ experience from the five matches. In addition, after each of those matches, the subjects participated in a short semi-structured interview to help the researchers to gather relevant information about the match, such as the strategies and techniques they used to play, if they felt physically or mentally tired or fatigued, if they had difficulties in playing or concentrating, and other general observations. The last interview of the session was longer for the subjects to further elaborate on their subjective views on the game experience, playing strategies and general open-ended opinions on the game. The GEQ instrument was applied once at the end of all matches due to its length (67 items in total, including all three modules) and the relatively short duration of each match. The duration of each experimental session was 37 min on average. Figure 3 represents the protocol for this procedure.

3.6. Design

The experiment has a between-subject design, as each user is assigned to a group and each group tested for only one task. The independent variable is the interaction mode with the game, with two levels (collaborative and competitive). The dependent variables are:

- The average attention of the participants (1–100);
- Their scores for the three GEQ forms (0–4).

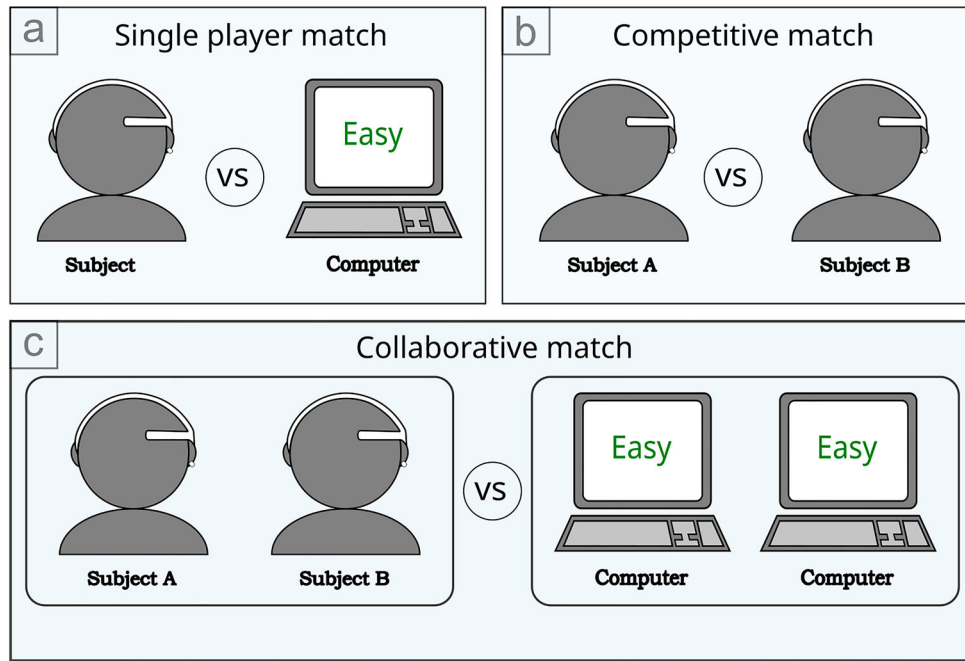


Figure 2. Schemes for (a) single player, (b) competitive and (c) collaborative matches.

The average attention values were collected at the end of each match to represent the subject's average performance during that match. The GEQ scores were all acquired at the end of the experiment session, as described in Section 3.5. Therefore, there are five average attention values and one score for each GEQ component per subject.

4. Results

The analysis of the data gathered during the experiment, both qualitative and quantitative, is presented in relation to each research question. Statistical tests were employed to analyse the dependent variables, i.e. the average level of attention from the subjects and the scores for the GEQ form. For all quantitative data, a D'Agostino & Pearson normality test was applied to verify the normal distribution of the values for the statistical analyses. All tests were performed using partial eta squared (η_p^2) as measure of effect size.

4.1. RQ1: Is there a difference in performance between the collaborative and the competitive groups?

To address this research question, an independent samples t-test was applied to compare the average score of the five matches that each of the eight subjects of each group played. The test revealed that there is no significant statistical difference between the average performance of both groups, $t(14) = 0.239$, $p = .814$. Table 1 presents the average attention scores of all matches from both groups.

In order to further investigate the differences between the two groups, the data were also analysed in terms of the individual matches, i.e. if there are any differences in the performance based on the number of matches played by each subject from both groups. As each match was played consecutively, a repeated-measures two-way ANOVA was applied by considering the matches as a within-subject time factor and the groups

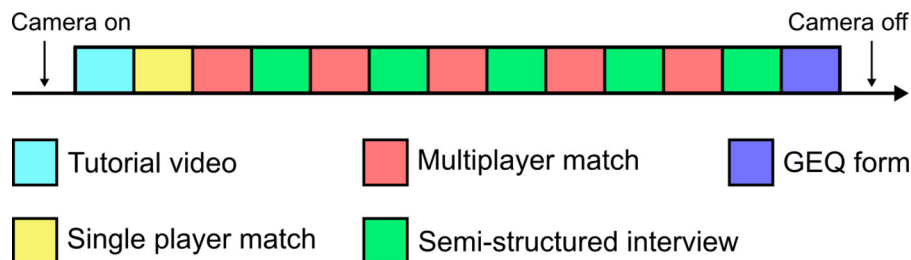


Figure 3. Experiment protocol (the lengths of the boxes are not proportional to each activity's duration).

Table 1. Statistics of average attention for all matches from both groups.

Statistic	Competitive matches					Collaborative matches				
	M1	M2	M3	M4	M5	M1	M2	M3	M4	M5
Minimum	50.2	42.3	42.8	37.7	38.6	35.6	27.2	30.2	36.4	27.6
25% Percentile	50.4	47.3	50.6	47.0	45.6	48.3	42.7	46.2	43.0	35.1
Median	56.1	52.9	57.7	51.4	50.9	61.5	61.7	58.3	50.2	45.2
75% Percentile	58.5	57.9	61.0	68.2	56.3	71.8	63.7	64.2	60.6	56.6
Maximum	74.3	68.4	70.8	84.7	67.3	88.3	72.3	71.9	74.0	67.5
Mean	56.8	53.4	56.5	56.8	51.6	61.0	54.1	54.7	52.6	46.7
Std. deviation	7.94	8.90	9.02	15.6	9.45	18.5	16.6	14.0	13.0	14.8
Avg. duration	74.8	58.0	54.8	54.0	48.5	42.5	42.8	69.3	85.8	36.8

as the between-subject treatment factor. This allowed analysing not only the differences between the groups, but the influence of the number of matches that each subject played on his/her average attention (e.g. fatigue and learning effects) and whether this effect is the same for both groups. Therefore, the independent variables are the group condition, with two levels (competitive and collaborative), and the time, with five levels (i.e. the matches, from the first to the fifth). The dependent variable is the average attention score.

Mauchly's sphericity test for the repeated measures is shown in Table 2. The main effect of time (matches) significantly violated the sphericity assumption, as the significance value is lower than 0.05 for the average attention, $W = 0.248$, $\chi^2(9) = 17.289$, $p = .046$. Therefore, the degrees of freedom for the main effect of time and its interaction with the between-group variable were corrected using Greenhouse-Geisser estimates of sphericity ($\epsilon = 0.656$). Levene's test for equality of variances was non-significant for matches M3, M4 and M5, but significant for matches M1 ($p = .034$) and M2 ($p = .042$). The choice of using the RM two-way ANOVA was kept nevertheless, as it is fairly robust when considering equal samples sizes and a relatively high significance value.

For the within-subjects effects, the test revealed that there is no significant main effect of time, $F(2.62, 36.7) = 2.17$, $p = .116$, $\eta_p^2 = .134$. This result suggest that, when considering only the time and ignoring the group factor, the number of matches that the subjects played does not have a significant effect on their average attention score. Table 3 shows the statistics for all matches of both groups, including the average duration of each match. The data from the Table and the large observed effect size suggest that the subjects may suffer from possible fatigue effects and tend to lose performance with consecutive matches, although this effect was not statistically significant.

For the between-subjects effect, the test presented the same result as the independent samples t-test, as expected, $F(1, 14) = 0.057$, $p = .814$, $\eta_p^2 = .004$. This suggests that there is no difference in performance for both groups, i.e. subjects playing competitively had the same performance of subjects playing collaboratively. Finally, the mixed within-between subjects relation of time vs. condition also shows no statistical significant difference, $F(2.62, 36.7) = 0.611$, $p = .591$, $\eta_p^2 = .042$. This means that, considering both factors simultaneously, there is no difference between the average attention of both groups in relation to time, i.e. the effects of time are the same for both groups. Figure 4 presents the average attention per match of all subjects and separated by mode of interaction. The figure corroborates with the results, as the average attention indeed slightly decreases over time, but this decrease is virtually the same for both groups.

There was also no significant difference in the average duration of the matches when considering the main effect of time $F(2.30, 32.1) = 1.22$, $p = .312$, $\eta_p^2 = .080$, nor the main effect of groups $F(1, 14) = 0.099$, $p = .758$, $\eta_p^2 = .007$. The graphs in Figure 5 show the average duration (in seconds) per match and the average duration considering both groups separately.

4.2. RQ2: Is there a difference in the game experience between the collaborative and the competitive groups?

As stated in Section 3.5, at the end of the session each subject was asked to fill out a GEQ form, which measures various aspects of the game experience and will help to answer this research question. The GEQ was chosen for this analysis as it not only provides insights regarding the overall game experience during and after the game,

Table 2. Mauchly's test for sphericity.

Within-subjects effect	Mauchly's W	Approx. χ^2	df	Sig.	Epsilon		
					Greenhouse-Geisser	Huynh-Feldt	Lower-Bound
Time	0.248	17.289	9	.046	0.656	0.878	0.250

Table 3. Statistics of average attention and duration (in seconds) from all matches.

Statistic	Match				
	M1	M2	M3	M4	M5
Minimum	35.6	27.2	30.2	36.4	27.6
25% Percentile	50.4	45.3	47.7	43.9	41.9
Median	57.8	55.6	57.7	51.4	48.4
75% Percentile	65.7	62.0	63.7	66.2	56.3
Maximum	88.3	72.3	71.9	84.7	67.5
Mean	58.9	53.8	55.6	54.7	49.2
Std. deviation	13.9	12.9	11.4	14.1	12.2
Avg. duration	58.6	50.4	62.0	69.9	42.6

but it also has a specific module designed to analyse the social presence of the game, i.e. how do players perceive and feel about the social interaction while playing the game. The results will be presented by comparing the scores, which ranges from zero to four, for each component of the core, social presence, and post-game modules of the GEQ. The highest values between both groups for each component are highlighted in bold. The scores were also compared between groups using a two-tailed independent-samples t-test.

The core module measures seven components: Competence, Sensory and Imaginative Immersion, Flow, Tension/Annoyance, Challenge, Negative Affect, and Positive Affect. In general, the competitive group had higher scores in most components, except for flow, challenge, and negative affect. However, none of these differences were statistically significant. Table 4 gives the average points in each component for both groups.

The social presence module measures three components: Empathy, Negative Feelings, and Behavioral Involvement. All scores were higher for the competitive

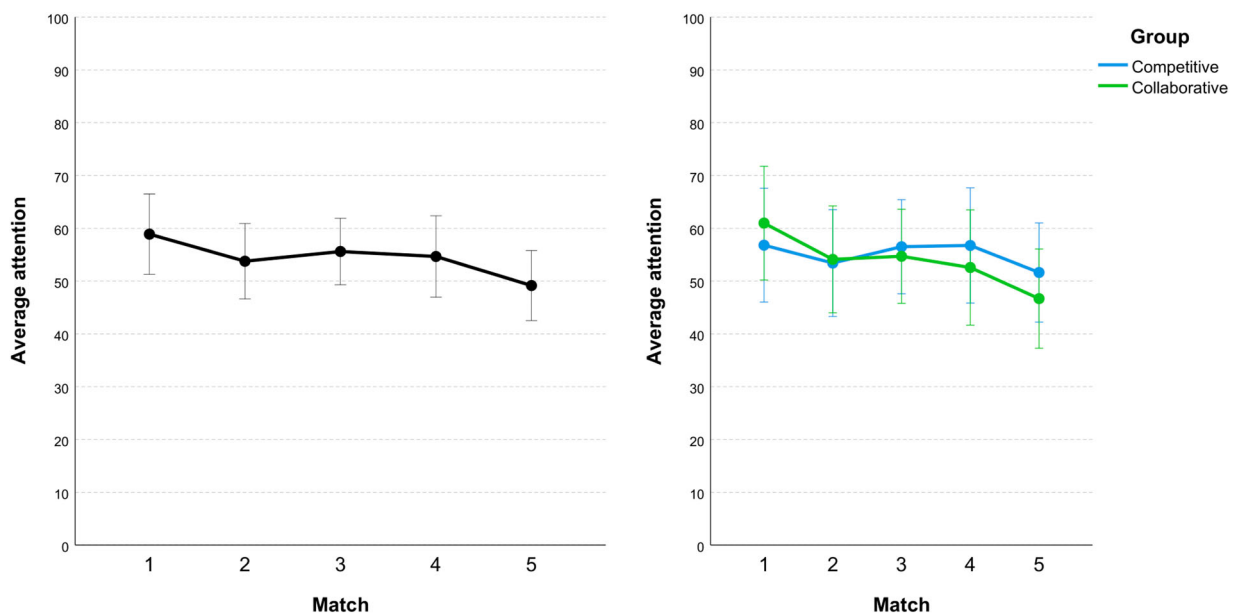
group. However, the Behavioral Involvement component had almost no difference between the groups. The Negative Feelings component had the highest difference, with 0.73 points, but again not statistically significant. Table 5 gives the average points in each component for this module.

Finally, the post-game module has four components: Positive Experience, Negative Experience, Tiredness, and Returning to Reality. All components were higher for the competitive groups, with exception of the Negative Experience component. The differences were also not statistically significant. Table 6 shows the average points of this module.

Based on the values of each module, it appears that the subjects of the competitive group had an overall slightly better experience playing the game. The differences in some key components were much higher, with over half-point of difference in the Competence, Positive Affect, Sensory and Imaginative Immersion, Empathy, and Positive Experience. Also, the value for the Negative Affect and Negative Experience are lower than the collaborative group. However, the Negative Feelings component has a 0.73 points higher value in the competitive group in comparison to the collaborative group.

4.3. RQ3: Is there a relation between performance and game experience in both groups?

The dependent variables were analysed in terms of their correlation with one another. The attention scores and each component of the GEQ modules were compared

**Figure 4.** Average attention per match for all subjects (left) and for both groups (right), with 95% confidence interval.

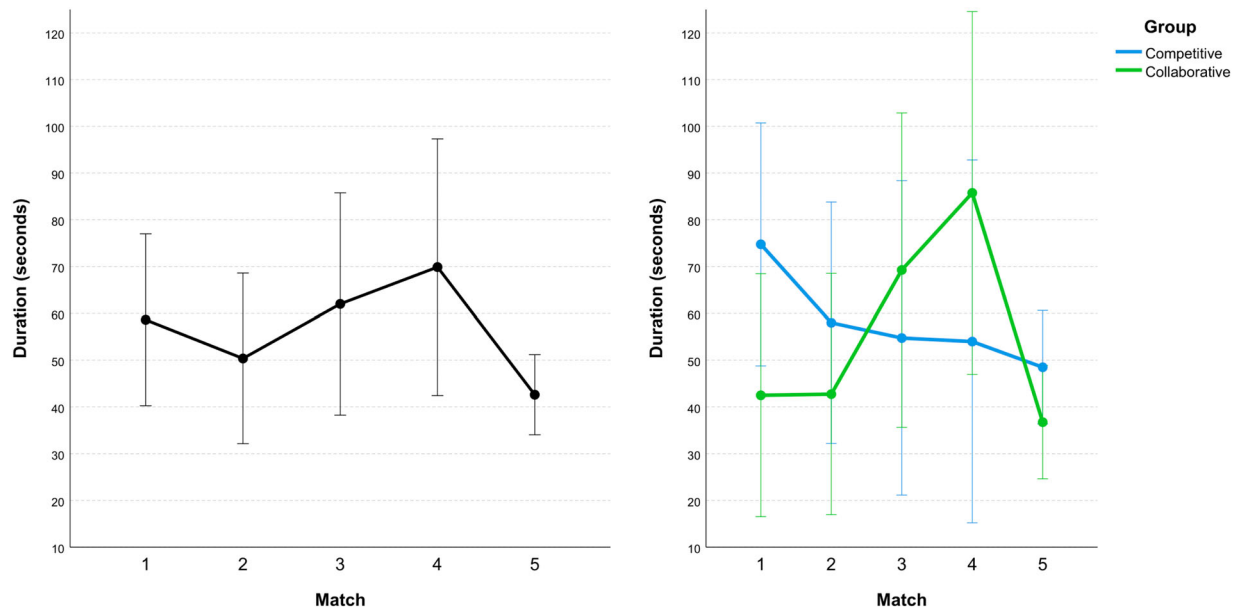


Figure 5. Average duration per match for all subjects (left), and for both groups (right), with 95% confidence interval.

Table 4. Average scores of the GEQ's core module components for both groups.

Component	Competitive	Collaborative	Difference	Sig.
Competence	2.13	1.60	0.53	.212
Sensory and Imaginative Immersion	2.25	1.92	0.33	.467
Flow	2.08	2.10	0.02	.945
Tension/Annoyance	0.71	0.54	0.17	.551
Challenge	1.75	1.88	0.13	.773
Negative affect	0.88	1.06	0.18	.567
Positive affect	2.68	2.28	0.40	.371

in regard to their correlation with each other's score. Table 7 shows the Pearson correlations of each statistically significant result. Each component in the first column is numbered and represented by its corresponding number in the first line.

The analysis shows interesting results, although some were expected. For example, the attention score is positively correlated to the Competence component of the core module, showing that the subjects felt more competent when their performance is high. Similarly, the Competence component is highly correlated with the Positive Experience component. In contrast, the attention score is negatively correlated with the Negative Experience component of the post-game module,

Table 5. Average scores of the GEQ's social presence module components for both groups.

Component	Competitive	Collaborative	Difference	Sig.
Empathy	2.19	1.60	0.58	.256
Negative Feelings	1.68	0.95	0.73	.101
Behavioral Involvement	1.48	1.46	0.02	.969

showing that subjects with low performance had a more negative experience with the game. Both scenarios were expected due to the nature of the task. Other examples of expected correlations were the positive correlations between the Negative Experience and both the Tension/Annoyance and Negative Affect components, and between the Positive Experience and Flow components. However, some correlations were unexpected, e.g. the Negative Feelings component is positively correlated with Sensory and Imaginative Immersion, Behavioral Involvement, Positive Affect and Empathy, being the latter two usually related to positive feelings.

4.4. RQ4: What are the subjective or qualitative differences in game experience between groups?

The qualitative data, obtained from the videos, observations and interviews, allowed analysing the results from the point of view of the subjects. This analysis may help to understand the differences in the game experience from both groups and reveal details regarding the BCI interaction with the game platform.

After all game matches, the subjects were asked whether playing the other game mode,

Table 6. Average scores of the GEQ's post-game module components for both groups.

Component	Competitive	Collaborative	Difference	Sig.
Positive experience	1.83	1.23	0.60	.245
Negative experience	0.33	0.50	0.17	.575
Tiredness	0.81	0.56	0.25	.659
Returning to reality	0.67	0.33	0.33	.399

Table 7. Pearson correlations among attention and GEQ scores.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1. Attention	–													
2. Competence	.666**	–												
3. Sensory and Imaginative Immersion	ns	ns	–											
4. Flow	ns	ns	.738**	–										
5. Tension/Annoyance	ns	ns	ns	ns	–									
6. Challenge	ns	ns	ns	ns	ns	–								
7. Negative affect	ns	ns	ns	ns	.736**	ns	–							
8. Positive affect	ns	ns	.701**	.606*	ns	ns	ns	–						
9. Empathy	ns	ns	.819**	.689**	ns	ns	ns	.725**	–					
10. Negative Feelings	ns	ns	.605*	ns	ns	.534*	ns	.536*	.619*	–				
11. Behavioral Involvement	ns	ns	ns	ns	ns	ns	ns	ns	ns	.574*	–			
12. Positive Experience	ns	.809**	.551*	.635**	ns	ns	ns	.649**	ns	ns	ns	–		
13. Negative Experience	–.556*	ns	ns	ns	.527*	ns	.692**	ns	ns	ns	ns	ns	–	
14. Tiredness	ns	ns	ns	ns	ns	ns	ns	ns	ns	ns	ns	ns	ns	–
15. Returning to Reality	ns	ns	ns	ns	.517*	ns	ns	ns	ns	ns	ns	ns	ns	.719**

ns, non-significant correlation. *Correlation is significant at the .05 level (2-tailed). **Correlation is significant at the .01 level (2-tailed).

i.e. competitively or collaboratively, would have been better or worse, both for the performance and the game experience. For the subjects in the competitive group, seven subjects stated that their performance would probably be better in the collaborative mode. Only one subject thought otherwise: *‘it (playing collaboratively) would probably be more difficult, as the two of us would have to synchronise our mental state in order to win, as opposed to concentrating individually. I think playing alone is easier in this sense’*. (S1, competitive group). In addition, four subjects stated that the game would probably be more fun in the other mode.

For the collaborative group, six subjects stated that their performance would probably be better in the competitive mode; the same six subjects also stated that the game experience would also be better, as the match would be fairer and they would be more excited to play against other player. Subject S2, for example, stated: *‘If I were playing against him (S4), I would be more excited to play and would try to concentrate more to defeat him. It also would probably be more difficult than to play against the computer’*. (S2, collaborative group). Subjects S6 and S8 thought that the game was hard against the computer: *‘I believe that the match would be more balanced if we were playing against each other, since playing against the computer was pretty hard. It felt like the computer team always had a higher attention than us, so playing against another person would feel more natural and humane, as our attention would oscillate more during the match’*. (S8, collaborative group)

In this context, the results could have been different if the collaborative group were to play against another pair of players. However, as pointed by Nijholt (2015a), in the context of HCI, one of the partners involved in a competitive or collaborative BCI-based game can be an artificial agent, playing and behaving in an human-like way. Albeit the pairs of subjects in

the collaborative group played against a pair of computer players, the core task of the activity remained the same. Although it is not known whether playing against another pair of humans would provide different results, isolating the collaborative aspect of the game from the competitive one (i.e. by having the subjects playing only collaboratively or competitively) was important to reduce possible confounding variables in the results, given the controlled design of the experiment.

Regarding the game strategies employed by the subjects, most of them were based on visually focussing on a particular point in the scenario. Other strategies included trying to remember details of a song, such as its rhythm or lyrics; focussing on controlling the breathing; perform mathematical operations; counting; and closing the eyes and imagining a specific scenario or memory. Some subjects also employed specific personal strategies, e.g. subject S16, who covered the neurofeedback interface element of the game, which is an indicator of the current attention of the player and is updated every second. The subject commented that *‘The constant movement of the bar (the attention indicator on the game interface) was hindering my focus, so I decided to just cover it with my hands so I could concentrate more’*. (S16, collaborative group). Subject S2, on the other hand, tried to visually concentrate on the same indicator, by trying to make it move towards the maximum value.

Two subjects, i.e. subjects S2 and S5, reported that wearing the headset device was causing discomfort due to its adjustment in the head. To maintain a stable quality of the collected signals, it is required that the device stays at a fixed position, limiting the subject’s head movements and applying a constant pressure on his/her right temple for physical support, which can cause discomfort for some users. Also, when signals of poor quality are received from the device due to

interference or misplacement, the game ceases to update the attention values in order to prevent faulty data in the quantitative analysis. In the majority of cases, the occurrence of such bad quality signals did not interfere with the game, as the BCI device itself adjusts the capturing of the signals within a few seconds; however, there had been two cases, both with subjects S13 and S15, in which the match was restarted due to the persistence of poor quality signals. It is not expected that such cases had interfered with the results, as they occurred within the first fifteen seconds of the match start.

In regard to the correlation analysis, Table 7 shows that there were several correlations between the GEQ scores from different modules. The correlation between attention and Competence, for example, shows that the subjects had a greater sensation of competence when their performance was high, while the negative correlation between attention and Negative Experience shows the inverse: when the performance was low, the user has a more negative experience with the game. These correlations, although some are expected, indicate that the GEQ was a reliable instrument to measure the subjective game experience from the participants in this experiment design. The correlations between Negative Experience with Tension/Annoyance and Negative Affect, for example, are expected due to the negative aspect of each component. It is interesting to notice that the Sensory and Imaginative Immersion component is one of the highest scoring components of all modules, showing that the neurofeedback aspect of the game can indeed increase its immersion, while also showing a positive correlation with other positive components, such as Positive Affect, Empathy and Positive Experience – although it is also correlated with the Negative Feelings component.

It is also interesting to notice how the Flow component is positively correlated only to other positive components, such as Positive Affect, Immersion, Empathy, and Positive Experience. Flow can be defined as a state of mind of complete immersion in a given activity, in which the person becomes more focussed and acts with total involvement in the task (Csíkszentmihályi 1975). Given this definition, it makes sense that a positive self-assessed value of Flow is highly correlated with Immersion and Positive Experience, for example. This result shows that, in addition to increasing the game immersion, the neurofeedback aspect of the game can also increase its flow (although this study cannot confirm a causal effect between the two). In the work of Nijholt, Bos, and Reuderink (2009), the authors also point to the possibility of using BCIs in order to not only promote a better reading of the flow experience, but also to adapt the game's challenges and

difficulty according to the user's current mental state, allowing one to achieve this state of flow more easily. In addition, the results of the experiment performed by Mladenović et al. (2017) showed that adapting the task difficulty increased the users' flow state in an EEG-based BCI, corroborating with the assumption that BCIs can be used to achieve higher levels of flow.

4.5. Additional analyses

This section describes additional results and comparisons that may complement the analyses from the research questions. Two additional analyses were performed: a comparison with the objective results of the multiplayer mode from the current experiment with the results from an experiment using single player matches, and a comparative analysis with results from literature studies that investigated the differences of competitive and collaborative interaction in games.

4.5.1. Comparison with single player mode

For a more comprehensive insight on the possible differences in gameplay, the multiplayer mode was also compared with the single player mode of the same game analysed in a previous experiment (Vasiljevic and de Miranda 2019b). This previous experiment had eight subjects playing four matches per day, during six days, in different game modes. For comparative purposes, only the matches from the first session of the experiment were analysed, since the subjects were in similar conditions (playing for the first time in the Easy difficulty).

The performance and duration were compared while considering the different groups (single player, collaborative, competitive) and different modes (single player, multiplayer). In both cases, there were no significant differences in performance between the subjects in single or multiplayer matches. The boxplots in Figure 6 show the average attention considering modes and groups.

As for the duration, there was a significant difference between the average duration of matches in single and multiplayer modes $F(1, 22) = 12.66$, $p = .002$, $\eta_p^2 = 0.365$. As expected, there was also a significant difference when considering groups $F(2, 20) = 5.62$, $p = .012$, $\eta_p^2 = 0.360$, with *post hoc* tests showing the differences between single player and competitive groups ($p = .031$) and single player and collaborative groups ($p = .022$). It is important to highlight, however, that the single player matches were played by considering the total average attention of the player throughout the match, instead of the rolling 20-s average that was employed in the multiplayer matches. The boxplots in

Figure 7 show the average duration between these modes and groups.

4.5.2. Comparative analysis

As commented on Sections 1 and 2, past studies also investigated similar aspects of multiplayer game interaction, although only a few were focussed on BCI-based games, and fewer compared competitive and collaborative modes. This section provides a brief overview of how the results of these works are compared to the results obtained in the present study. We also included regular non-BCI games in this analysis, since they may share characteristics that are independent from BCI controls.

Tables 8 and 9 present a summary of the experimental setup and results obtained from these related studies. We have included studies that focussed on the effects of multi-user interaction while playing a game, especially those using measurements of attention, since it was the employed BCI control signal in the current study, and game experience. Although a number of studies present results focussed on competitive or collaborative modes, we included only those that compared both modes directly.

As can be seen from the results shown in Table 8, there is a relatively small number of studies, to our knowledge, that directly compares both modes of interaction, and their results seems to converge in the same direction. In general, there are no significant differences in performance between playing competitively or collaboratively, nor when considering other metrics that may influence game experience (e.g. individual preference, task load or motivation). Our study presented similar

findings, since there appears to have no noticeable difference between both groups. It is important to notice that these similar results were obtained using different control signals, performance metrics, game genres, qualitative instruments and game mechanics, which indicates that mode of interaction have little impact in game experience, or that any influence is mostly due to personal preference or other qualitative factors. According to our qualitative analysis, for example, individual subjects may prefer to play competitively or collaboratively, and this preference could possibly increase motivation to play (and thus affect individual performance and game experience to some level), although having small practical effects on the performance.

Regular, non-BCI games still contribute to this analysis, in the sense that they may share factors that are independent of BCI controls. It can be noticed, for example, that there is usually no significant differences in objective metrics in relation to game modes, although there are differences in other subjective factors, such as friendship quality and motivation. This may indicate that the mode of interaction tends to influence on the neurological or psychological level, as there are physiological factors, such as EEG asymmetry, that varies according to interaction style. Other results, such as the higher satisfaction for competitive play compared to collaborative play, is also in accordance with the results obtained in the present study. Although not statistically significant, the results from the GEQ scores, combined with the qualitative analysis, suggest that there was an overall better game experience in playing competitively.

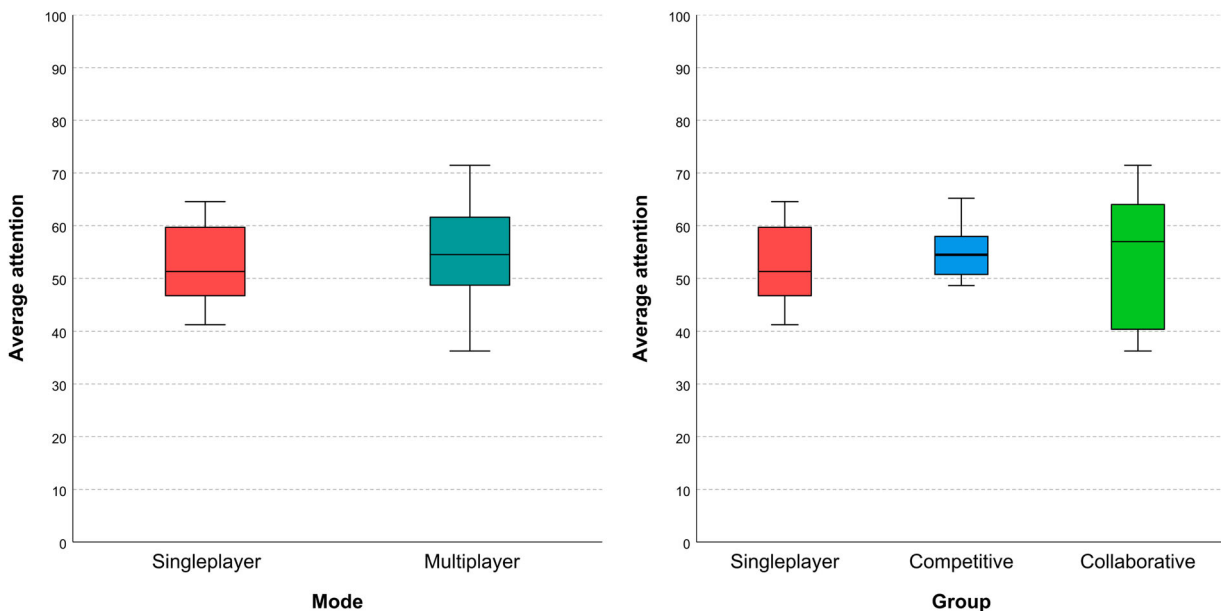


Figure 6. Average attention per mode (left) and per groups (right).

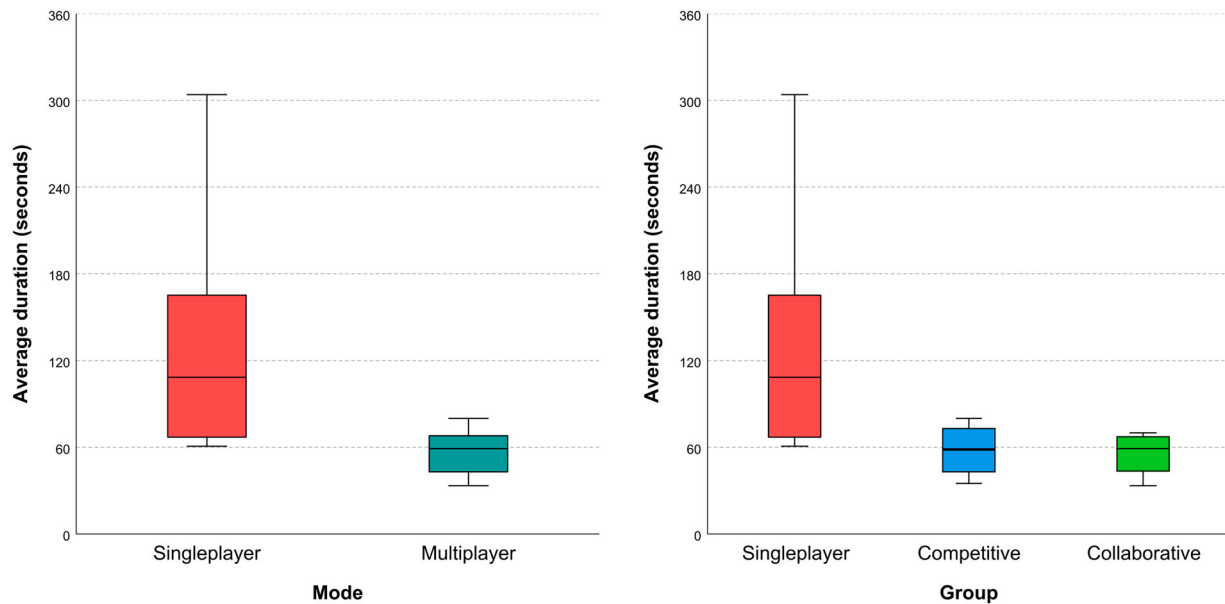


Figure 7. Average match duration per mode (left) and per groups (right).

Table 8. Summary of BCI studies that compared competitive, collaborative and/or single player modes of playing.

Reference	Control signal	Comparisons	Summary of results
Bonnet, Lotte, and Lécuyer (2013)	Motor imagery	Single player x Collaborative	Multiplayer interaction is significantly preferred over single player mode. No significant difference in preference and performance for competitive and collaborative conditions
Ling and Vucković (2016)	Attention, relaxation	Competitive x Collaborative	No significant difference in performance between playing collaboratively or competitively
Luca et al. (2021)	α power	Competitive x Collaborative	High brain connectivity for both modes, but lower for competitive in comparison to collaborative mode. No significant difference in task load for both modes
Sekhavat (2020)	Attention	Competitive x Collaborative	No significant difference in attention, motivation and game experience between playing in both modes

Labels: Single player. Competitive. Collaborative.

As noted in the work of ter Vrugte et al. (2015), differences in conditions may exist for different subgroups of subjects; for example, high-performing players may feel more confident or motivated to play in competitive conditions, without depending on the performance of the other player. The domain of application, game genre, age of subjects and other social and technical factors may also have influence on this difference.

5. Discussion

The discussion is based on the experiments' results and the research questions, in addition to the limitations and threats to the validity of the study and additional comments based on the qualitative experience of the subjects in relation to the game.

The RQs are discussed regarding both the qualitative and quantitative data. Although the first RQ is

Table 9. Summary of non-BCI studies that compared competitive, collaborative and/or single player modes of playing.

Reference	Comparisons	Summary of results
Lin et al. (2017)	Competitive x Collaborative	No significant difference in learning performance and motivation between both modes. Significantly higher satisfaction for playing competitively in comparison to playing collaboratively
Peng and Hsieh (2012)	Competitive x Collaborative	No significant difference in performance between playing collaboratively or competitively. Significantly higher motivation and goal commitment for playing collaboratively in comparison to playing competitively
Salminen et al. (2009)	Competitive x Collaborative	No significant difference in frontal EEG α asymmetry between playing collaboratively or competitively
Verheijen, Stoltz, and Cillessen (2019)	Single player x Competitive x Collaborative	Significant changes in behaviour between playing competitively or playing cooperatively. Higher positive behaviour in both modes; lower friendship quality after playing competitively in comparison to single player, and more positive behaviour while playing cooperatively
ter Vrugte et al. (2015)	Competitive x Collaborative	No significant difference in learning effects between playing collaboratively or competitively. Positive effects of competition for above-average participants, but negative effects for below-average ones in specific situations

Labels: Single player. Competitive. Collaborative.

fundamentally answered by quantitative analysis, the second RQ may require additional qualitative analysis due to its subjective nature, i.e. its result may be influenced, or corroborated, by the subjects' opinions and perceptions. Additionally, the subjective analysis may help to explain the quantitative results obtained from both RQs.

For the first research question, the quantitative results showed that there was no significant difference between the performances of both groups. Given the average attention values throughout the matches (Figure 4), both groups had indeed very similar performances. The decrease in the average attention after consecutive matches was somehow expected given that there is no fixed interval time between each match. Each pair of subjects was free to start the next match as soon as both subjects were ready. The highest difference was from the first match in comparison to the fifth, although there is a small difference starting from the second match. With additional data, this could indicate a possible threshold point in which the average person can no longer play a neurofeedback game consecutively without loss of performance, and could be used as a reference for future experiments. This also means that, for a game designer, it may be interesting to develop mechanisms that prevent the player from immediately starting a match after the last one is over – for example, animations or extra tasks after the game.

In regard to the second research question, the quantitative results based on the GEQ scores have shown that there was a moderate difference in the game experience between the groups, with the majority of scores higher for the competitive group. For the core module, which measures the overall game experience, the collaborative group only scored higher for the Challenge and Negative Affect components, albeit the differences were negligible (lesser than 0.10) for Tension/Annoyance and Challenge components. The scores with higher differences indicate that the competitive group had a slightly better experience, especially taking into consideration that the Positive Affect had the higher difference between the groups, while the collaborative group scored higher for the Negative Affect component.

The scores from the social presence module were also higher for the competitive group, although the highest difference (0.73 point) was in the Negative Feelings component. Interestingly, this higher score could be expected due to the nature of the competitive mode, as the success of the one player depends on the failure of the other. This could induce the losing player to direct his/her frustration and other negative feelings to his/her opponent or to the game itself. However, the same could happen in the collaborative group, as the

members of the losing team could blame their respective partner for the loss, or feel guilty or responsible for the defeat. A bigger sample size and a larger study focussed on this specific interaction could provide interesting results. It is also worth noticing that the Empathy component (which measures the capacity to understand or feel what the other person was experiencing from his/her perspective) was also higher for the competitive group. One could expect, given the score from the Negative Feelings component, that the collaborative group had a higher score for this component. This result, however, is in accordance with the findings of Spapé et al. (2013), in which the matches played in the most competitively manner had a higher score for the Negative Feelings component in the social presence module in comparison to the cooperative matches.

The post-game module also gave interesting results, as the scores also indicate that the players from the competitive group had an overall better experience – specifically, the scores from the Positive Experience and Negative Experience, both favouring the competitive group. Nevertheless, the competitive group also had a slightly higher score for the Tiredness component, which is not necessarily a positive factor. Based on the comments from the subjects, acquired during the interviews, it appears that there is a greater pressure to keep a high average attention while playing against another player, as it will directly affect the outcome of the match. In the collaborative group, however, having a slight decrease in the attention does not significantly impact on the performance of the group, as the value is also averaged with the other player's score.

It is important to highlight, however, that although the competitive group had an overall slightly better game experience, it does not mean that the collaborative group had, in general, a negative experience. The scores from the Negative Experience component, in addition to the data from the interviews, suggest that the subjects enjoyed the game and the experience, as its value, in average, was the lowest among all components from all modules.

Both the quantitative and the qualitative results appear to be in accordance to other similar studies from the literature. The work of Sekhavat (2020), for example, which employed the same instruments for measuring performance and game experience as the present study (although with a different study focus and game genre), also found no statistical difference between competitive and collaborative conditions in both measures, even considering a higher sample size. Salminen et al. (2009) and Bonnet, Lotte, and Lécuyer (2013) also found no difference in performance between competitive and collaborative conditions in their games;

however, in all of these studies, the subjects preferred the competitive/collaborative conditions in relation to single-player conditions, even when their performance was worse.

The combination of these results could indicate possible directions for the design and implementation of future feedback-based BCI games. The data obtained from the current experiment and past results from the literature suggest that multiplayer modes are generally preferable over single player ones, independently of being competitive or collaborative. This also indicates that the preference over one mode or the other is largely personal, as shown by the qualitative results. Thus, when possible, both modes of interaction could be made available without loss of performance and without largely impacting the game experience. Most subjects in our study, however, preferred to play or would rather have played competitively instead of collaboratively. The task of concentrating as a game mechanic may influence in this decision, since focussing one's attention is naturally an individual task.

In this particular game, the attention values of both players are averaged and compared to a threshold (the opposite teams' attention). Although the game provides a visual indicative of each player's individual attention, their attention metric is shared in the game UI. Despite being a seemingly minor detail, a separate attention bar for each player or a visual alert for when one of the players are performing poorly in comparison to the other could have yielded different results, since there would be more pressure to increase one's performance individually. In addition, playing against a team of other human players could also have impacted game experience, as some subjects commented that the gameplay could feel more natural in comparison to playing with the computer players.

Although there was a significant difference in the average match duration between single player and multiplayer matches, the employment of the 20-s rolling average instead of the total attention values could be responsible for this difference; therefore, it cannot be concluded that the game mode alone influenced the match duration. Performance-wise, however, this change does not appear to have influence in the average attention, since there was no difference in performance between playing in single or multiplayer modes. Nevertheless, from a qualitative perspective, long matches could have greater fatigue effects as the player must focus for longer, therefore hindering the game experience.

Smaller time windows for the scrolling average could result in even shorter matches, although the attention values would have higher variance and may appear random to the player, since its value is updated roughly

every one second. The choice of a 20 s time window, obtained empirically, appear to offer balanced results, with an average of approximately one-minute matches for the easy difficulty in this particular game. When considering harder difficulties, however, one should take into consideration that the player would have to concentrate more for the same duration, as aforementioned. In this case, the adjustment of the difficulty could be done either in the duration of the match (concentrating steadily for longer periods of time) or the intensity of the concentration (concentrating more in shorter periods of time).

The employed control mechanics may also influence on the performance and game experience of the subjects. As mentioned in Section 3.4, for this current study, the average between the attention values of both subjects was employed during collaborative gameplay, while the absolute difference between the average attention of players was employed for the competitive mode. Considering related studies, other control strategies could produce different results. In the work of Luca et al. (2021), for example, players should maintain their relative α value (correlated with attention/relaxation using neurofeedback) above 10% of the opponents' value for scoring in the competitive mode, while they should maintain their relative α value in the range of 5% of their pair's value for scoring in the collaborative mode. As commented by the researchers, while inducing a higher synchrony between the participants' brain waves during collaborative mode in comparison to the competitive mode, it also induced the player with higher α value to reduce its performance to be in accordance with their pair, while in the competitive mode, the other player having a low α value was an advantage. In our study, players' performance scores are less dependent during collaborative mode, since a low performance player must focus on increasing its attention value for increasing the team's overall score. It could be interesting to investigate the combination of both strategies, i.e. group versus group performance, considering that both players in each team must maintain their attention scores in the same range of values, but also higher than the opponents' score.

Finally, it is also important to note that these results are based on an active use of the control signal. Passive gameplay using neurofeedback, in which the player does not actively concentrate on the control signal (e.g. the attention value being employed for controlling or adjusting indirect aspects of the game, such as its difficulty) could provide different results for performance. Based on the results of related works, it appears that the mode of operation (active or passive) does not have influence over the game experience of different

modes of interaction (collaborative or competitive), since the same conclusions were obtained regardless of this factor.

5.1. Limitations and threats to validity

As shown by the quantitative results, a high number of matches could hinder the subjects' performance over time. In this particular experiment, the number of matches was kept at five as the fatigue effect was also investigated. To minimise the effects of subject fatigue during future experiments, it would be prudent to conduct shorter sessions (e.g. three matches instead of five per session) with a fixed-duration interval between the matches. This also relates to the application of the GEQ form, which was applied only at the end of the five multiplayer matches. In future studies, a GEQ form could be applied at the end of each match, allowing for a better understanding of the players' perception of the game experience – Although a smaller number of matches would be recommended, as the form is relatively extensive. Also, the sample size (sixteen subjects, eight pairs) may be considered small for a comparative study. As the tests suggest, an increased number of participants could have led to statistically significant results, e.g. for the possible effect of the number of matches in the subjects' attention metric, especially considering the high effect sizes detected in some of those tests. In future studies, it would be interesting to have a bigger sample size to provide increased external validity and statistical power of the results.

It is also important to highlight the limitations of the employed BCI device. As the device has only one sensor positioned at location Fp1 (above the left eyebrow), the captured signals are subject to electrooculogram and electromyogram artefacts – i.e. contamination of the signals due to movements of the eye and face muscles (Fatourechi et al. 2007; Jeunet et al. 2018) –, which may affect BCI performance. The employment of more robust EEG equipment could provide a higher spatial resolution and reliability of the captured signals in future studies. It is worth noticing, however, that both the attention metric provided by the device and the quality of the signals have been evaluated and compared with medical-grade equipment for their reliability in scientific research, as stated in Section 3.3.

6. Conclusion

In this paper, the procedure and results of a controlled experiment with a multiplayer BCI-based computer game are described. The experiment was conducted with 16 subjects divided into two groups of eight

pairs, each playing five matches of the game in a different condition. This experiment aimed to investigate the influence of social interaction on both conditions, i.e. competitively and collaboratively, in the subjects' performance and game experience.

The experiment's results showed that, although there is no statistically significant difference in performance between the competitive and the collaborative groups, there are specific differences in the game experience from playing competitively and playing collaboratively. The results revealed that the players in the competitive groups had an overall slightly better experience, as the majority of metrics had a higher value in comparison to the collaborative group, but the differences were not statistically significant. Also, there is a small decrease in the players' performance from the first to the last matches of the sessions, indicating that playing many consecutive matches of attention-based games may cause fatigue effects and lower the average attention of the subject over time.

The performance and game duration of the multiplayer matches were also compared to single player matches. Although there was no difference in the players' performance, there was a significant reduction in the duration of the matches while playing in multiplayer mode. However, this change could be attributed to the scrolling average calculation of the attention values in contrast with a total average, and not necessarily to the mode of interaction.

These results corroborate with the data obtained from past and related studies, and could be employed as directions for the design and development of future neurofeedback-based games. As the mode of interaction had no influence on the players' performance or game experience in multiplayer modes, the design choice of employing competitive or collaborative modes depends on the intended design of the game, its purpose, and the target public, since the game experience in both modes appear to rely on the personal preference of the players. However, multiplayer modes are generally preferred over single player ones for this particular type of BCI game. Playing actively or passively also does not appear to influence on the game experience on both modes of interaction. In general, however, players in the collaborative group would rather have played competitively, and players in the competitive group would rather stay playing competitively. This may indicate that the mechanics of concentrating to play, being intrinsically individual, may also induce a more individual and competitive gameplay.

For future works, conducting a within-subjects longitudinal study could lead to interesting results,

as every subject would play the two game modes and compare the experience of both. The effects of time on the subject's performance and opinions throughout the sessions could also be analysed. Furthermore, it would also be interesting to investigate the effects of having groups of two or three subjects playing both competitively and collaboratively (i.e. group vs. group conditions).

Note

1. *Steady-state visually evoked potential*. A control signal in which a brain response is evoked based on a repetitive visual stimulus.

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